LLM-Based Course Comprehension Evaluator

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Abstract. Large language models (LLMs) like GPT-4 reshape intelligent tutoring systems by enabling nuanced natural language interactions. Leveraging LLMs' capabilities, this study introduces an innovative Lesson Comprehension Evaluator, utilizing advanced Natural Language Processing (NLP) methods and Augmented Retrieval Generation (RAG) to assess course material comprehension. Through a web interface, students engage with tailored questions and receive feedback, fostering immersive learning experiences. Each response undergoes rigorous evaluation against a ground truth LLM-generated knowledge base, encompassing semantic comprehension, specificity, and correctness metrics. These evaluations provide insights into students' course understanding, informing future pedagogical strategies. By incorporating auditory options for accessibility and gamification elements for enhanced engagement, this approach facilitates self-paced, deeper learning, fostering dynamic and enriching learning environments.

Keywords: Intelligent Tutoring Systems · GPT-4 · Retrieval Augmented Generation (RAG)

1 Introduction

In the realm of Intelligent Tutoring Systems (ITS), several significant advancements have been made in recent years, as evidenced by a series of papers published from 2019 to 2024. Explainable Artificial Intelligence was introduced in ITS to foster student engagement and understanding [1]. The integration of ITS with personalized learning (PL) gained traction [2]. Systematic literature reviews underscored the evolution of ITSs with natural language, showcasing various mechanisms for simulating human dialogue and pedagogical strategies [3, 4].

Papers explored the potential of LLMs in generating reading comprehension exercises [5] and developing adaptive practicing systems [6], highlighting their efficiency in question generation and mastery detection algorithms. However, challenges such as bias mitigation and evaluating multi-turn conversations remained focal points for future research [7]. Additionally, studies investigated the mathematical capabilities of LLMs and emphasized the need to address knowledge gaps and misconceptions in educational contexts [8, 9].

Innovative solutions, such as AI Tutors and Virtual Teaching Assistants (TA), showcased the integration of advanced AI technologies, including Large Language Models

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(LLMs), for personalized and adaptive learning experiences [10, 11]. These systems leveraged state-of-the-art LLMs, such as GPT-4, to provide accurate answers, tailored tutoring, and comprehensive feedback [11–13]. However, challenges such as scalability, interpretability, and ethical considerations remained pertinent [10, 11, 14].

In parallel, the research explored the authoring scaling of AutoTutors and the real-life usability of LLMs in science education [15, 16]. Studies on Retrieval Augmented Generation (RAG) systems emphasized the importance of careful design considerations and domain-specific fine-tuning [17, 18].

The transformative breakthroughs in NLP arrived with the ascent of deep learning methodologies and the development of LLMs, such as OpenAI's GPT [20] series and Google's BERT [21]. These LLMs are characterized by their massive scale, parameter counts, and ability to generate coherent and contextually relevant text.

The proliferation of LLMs has catalyzed a revolution in NLP, enabling unprecedented capabilities in tasks ranging from machine translation and question-answering to text summarization and dialogue generation. RAG is a pioneering paradigm within the domain of NLP. It integrates LLMs with external knowledge sources to leverage the retrieved knowledge to augment text generation capabilities.

This study introduces an innovative Course Comprehension Evaluator using LLMs (LLMCCE). The web interface allows students to engage with tailored questions and receive feedback, providing insights into their understanding and facilitating self-paced, deeper learning.

Subsequent sections concisely expose the technical underpinnings pertinent to the terminologies utilized. Finally, an exposition of the evaluation outcomes, which at the same time evaluate GPT-4 effectiveness in creating meaningful and on-target questions from textual content, derived from domain expert assessments and pilot study findings is presented, complemented by examining associated limitations and avenues for future research in this domain.

2 LLM-Based Course Comprehension Evaluator (LLMCCE)

The main modules that comprise our proposed approach are the User Interface, Timeline Creation Module, Evaluation Module, and Vector Database. A high-level architecture is shown in Fig. 1.

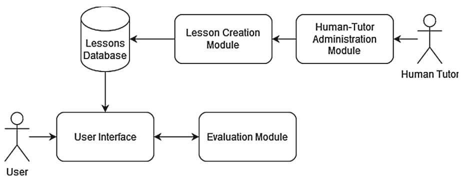

Fig. 1. High-level architecture of LLMCCE

2.1 User Interface Module

In general, several elements are included in the intelligent tutoring system's user interface to make learning easier and more comfortable. In our approach, one notable feature is the flexibility for students to choose the precise lesson file they want to practice. This feature empowers students to actively engage with the educational information in a way that is in line with their learning preferences and aims while also encouraging autonomy and agency among students and a sense of ownership over the learning process.

After the student chooses the desired lesson, in our case History, the main interaction screen changes and shows the corresponding timeline chosen.

Central to this new screen is the prominent display of the timeline, strategically positioned to facilitate student engagement and navigation through the educational material. The timeline is presented in a vertical orientation, commencing at the topmost portion of the screen and extending downwards, affording students a comprehensive overview of the chronological progression of events or learning milestones, as shown in Fig. 2.

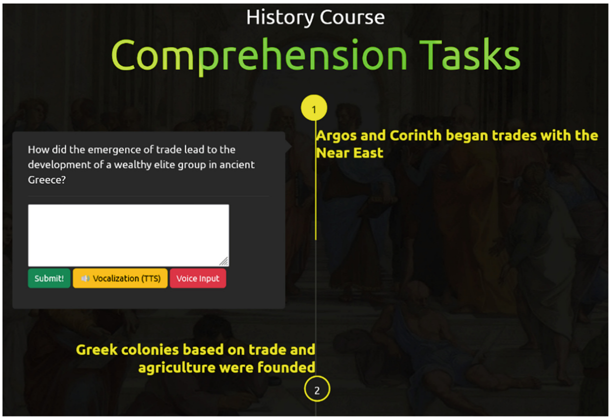

Fig. 2. The Timeline student interaction screen

2.2 Human Tutor Administration Module

In the administrative interface of the intelligent tutoring system, the human tutor is vested with significant autonomy and flexibility in shaping the pedagogical trajectory of the educational experience. A key feature of this interface empowers the human tutor to define the parameters governing the generation of a specific number of questions by the model or to directly input custom questions onto specific textual content, thereby circumventing the conventional Lesson Creation Module.

By bypassing the Lesson Creation Module, described in later sections, in favor of direct intervention by the human tutor, the system transitions into an evaluative role, wherein the focus shifts towards assessing learner comprehension and mastery of the educational content. The lesson is promptly saved to the database upon completion of the instructional design process, ensuring seamless integration into the overarching educational framework.

This approach not only expedites the content creation process but also affords the human tutor greater freedom and flexibility in tailoring pedagogical pathways to address learners' diverse needs and preferences. By emphasizing human expertise and input, the system fosters a collaborative educational environment wherein the complementary strengths of automated technology and human ingenuity converge to facilitate enriched learning experiences.

2.3 Lesson Creation Module

In creating lessons within the intelligent tutoring system, a structured procedure is followed to harness the capabilities of both human input and advanced natural language processing technologies. Initially, a human tutor accesses an administration panel designed for lesson creation, wherein they are prompted to insert textual content that forms the basis for constructing a timeline within the educational material. This textual content (TC) can range from concise excerpts to comprehensive passages, encompassing various educational resources such as book paragraphs, chapters, or entire books. Subsequently, upon submission of the inserted text, it undergoes processing within a Python script specifically engineered for this task.

Within the confines of the Python script, a meticulously crafted prompt is formulated, tailored to elicit pertinent information from the provided text and to guide subsequent question generation. This prompt encompasses instructions for extracting key points from the text, a predefined template for structuring responses, and the number of questions the tutor requested [NQ]. Leveraging the robust capabilities of the GPT-4 language model, this prompt is then deployed to generate responses that align with the specified template. The structure of this prompt is as follows:

"Create a timeline of [NQ] important events covering all of the given text. The presentation of the timeline will be presented without break lines and three keywords [referring_date, event, question] separated by comma as follows: referring_date: [Date], event: [Event], question: [Question] of understanding about the event. The keywords must be mentioned every time in the response. Text: [TC]".

The resultant responses generated by GPT-4 are subject to post-processing procedures, wherein the data is organized and stored in individual PHP files. A salient feature involves automatically generating PHP files each time a timeline is created, facilitating the storage and dissemination of educational content. This innovative functionality underscores a commitment to seamless collaboration and knowledge sharing among students and tutors within the educational ecosystem. A detailed low-level architecture of the lesson creation module is presented in Fig. 3.

This structured procedure underscores a symbiotic relationship between human expertise and artificial intelligence, wherein the computational power and natural language processing capabilities of state-of-the-art language models complement human tutors' nuanced understanding and instructional insights. By seamlessly integrating human input with advanced technological solutions, the intelligent tutoring system facilitates the creation of dynamic and engaging educational content tailored to learners'

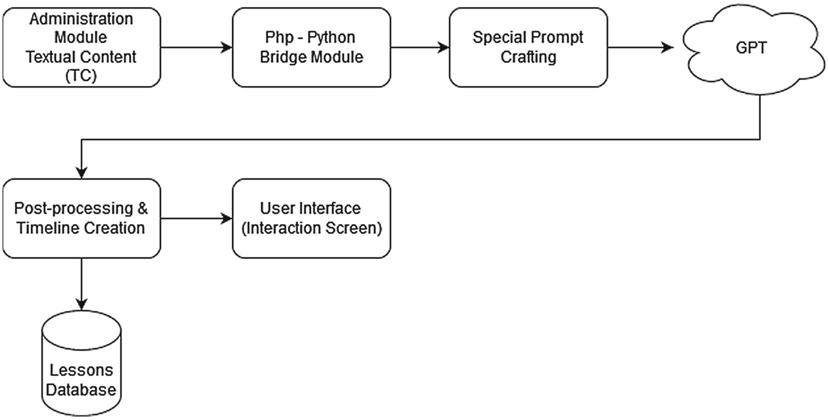

Fig. 3. Detailed low-level architecture of Lesson Creation Module.

diverse needs and preferences. Moreover, this procedure exemplifies a synergistic approach toward educational content creation, wherein the fusion of human creativity and machine intelligence culminates in developing pedagogically sound and intellectually stimulating learning experiences.

2.4 Evaluation Module

In the context of this step, the utilization of a vector database [22] plays a pivotal role in facilitating the storage and manipulation of textual data within the intelligent tutoring system. A vector database, in essence, is a specialized repository designed to store vectors, which are mathematical representations of data points in a multi-dimensional space. These vectors capture semantic relationships and contextual information inherent within the textual content, thereby enabling efficient retrieval and analysis of information.

Specifically, in our approach, the vector database serves as a repository for storing vectors generated from textual inputs provided by human tutors. Upon submission of a text by the human tutor to create a timeline within the educational material, the system initiates an auxiliary procedure. This procedure involves tokenizing the text and segmenting it into predefined chunk sizes, breaking the textual content into manageable units for processing.

Subsequently, each chunk of text transforms into a vector embedding utilizing an embedding model, with the specific model employed herein denoted as "text-embedding-ada-002" [20]. This embedding model facilitates the conversion of textual information into dense numerical representations, capturing the semantic nuances and contextual intricacies embedded within the text. Through this process, the textual content provided by the human tutor is encoded into vector embeddings, effectively encapsulating the semantic essence of the input within a mathematical framework.

This database is meticulously calibrated to operate using cosine similarity as its primary similarity metric, a mathematical measure widely acknowledged for its efficacy in capturing the degree of similarity between vectors in a multi-dimensional space. Furthermore, the dimensionality of vectors stored within the database is meticulously defined to adhere to the text embedding model's specifications. Specifically, the dimension size is set at 1536, aligning precisely with the dimensional requirements of the employed embedding model.

Completing this encoding process, the vectorized representations of the human tutor's text are inserted into the vector database for storage and subsequent retrieval within the system's operational workflow.

Within the architectural framework of our approach, incorporating a dynamic timeline feature serves as a cornerstone in facilitating interactive learning experiences.

Integral to this timeline are questions generated by the Lessons Creation Module, strategically positioned to prompt learner engagement and comprehension. Each question is complemented by a textarea input field, allowing students to articulate their responses effectively. To enhance accessibility and accommodate diverse learner needs, the system incorporates text-to-speech functionality, enabling auditory presentation of questions and automatic speech recognition (ASR) for seamless input through spoken responses.

Upon submission of a student's response (UR), the system initiates a comprehensive evaluation process orchestrated by a designated Python file. This evaluative procedure commences with a retrieval operation targeting the vector database to source relevant information pertinent to the question. Employing sophisticated retrieval algorithms, the system identifies the top three most relevant records from the vector database, contextualizing the subsequent evaluation process.

With relevant data, the system proceeds to solicit responses from the GPT-4 model, leveraging its advanced natural language processing capabilities. These responses, generated in response to the same question (UQ) presented to the student, serve as the benchmark ground truth (GT) against which student responses are evaluated. RAG's response is considered as the authoritative reference point for subsequent evaluations.

Subsequently, the system engages GPT-4 again, this time tasking it with the role of evaluator. A specialized prompt is crafted to guide the model in assessing the student's response vis-à-vis the generated ground truth, focusing on evaluating detail, semantic comprehension, and correctness. Each aspect is meticulously scrutinized, and a score ranging from 0 to 20 is assigned to quantify the quality and accuracy of the student's response relative to the established ground truth. The prompt follows:

"Act as a kind assistant, History Teacher Assistant. I want you to evaluate answers [UR] and [GT] against each other for question [UQ]. Taking [GT] as the correct reference, answer for [UR] the following: Detail:[Grade from 1 to 20], Semantic:[Grade from 1 to 20], Correctness: [Grade from 1 to 20]. Then average the three ratings and display Final Score:[Average from 1 to 20]".

The evaluation results are returned to the user interface. Additionally, a brief explanation of how GPT-4 evaluated the student's response is returned for better interpretability of the procedure.

2.5 LLMCCE Evaluation

In evaluating the correctness and effectiveness of the intelligent tutoring system's question generation and response capabilities, a rigorous assessment methodology is employed, leveraging the expertise of domain-specific professionals. This evaluative process involves soliciting feedback from three domain experts who possess specialized knowledge and proficiency in the subject matter under consideration. The experts evaluate questions and corresponding model-generated answers, assessing their relevance, accuracy, and alignment with the provided context.

The domain experts are provided with a standardized questionnaire comprising 15 question-answer pairs, each representing a distinct instance of model-generated content to structure the evaluation process. The experts are instructed to employ a classification method wherein they assign a score based on the question's relevance to the provided context and the adequacy of the model-generated answer in addressing the corresponding question. Specifically, a score of 2 is assigned if both the question and answer are deemed relevant and satisfactory, and a score of 1 is assigned if the question is relevant but the answer falls short of expectations. A score of 0 is reserved for instances where the question is deemed irrelevant to the context. This evaluation approach also evaluates how well an LLM such as GPT-4 can extract specific questions from a text and identify its main points.

Upon completion of the evaluation process, the results from each domain expert are compiled and analyzed to gauge the accuracy and effectiveness of the model's question generation and response capabilities. Notably, the evaluation results indicate a high degree of alignment between the model-generated content and the provided context, with the vast majority of question-answer pairs receiving scores of 2 or 1. Specifically, the first domain expert submits 13 pairs with a score of 2 and 2 pairs with a score of 1, while the second domain expert assesses 12 pairs with a score of 2 and 3 pairs with a score of 1. Remarkably, the third domain expert's evaluations mirror those of the first expert, further validating the consistency and reliability of the assessment process.

Crucially, none of the domain experts identify instances where the model-generated content deviates significantly from the provided context, as evidenced by the absence of scores of 0. This observation underscores the robustness and accuracy of the model in generating contextually relevant questions and responses (Table 1).

Table 1. The model generated Q&A pairs evaluation from domain experts.

Domain Experts	Domain Expert Evaluation		
	0	1	2
1^{st} DE	0	2	13
2^{nd} DE	0	3	12
3^{rd} DE	0	2	13

In aggregate, the evaluation results yield an overall accuracy rate of 84.4%, indicating a high level of proficiency and effectiveness in the system's question generation and response capabilities. This percentage is a key metric in assessing the system's performance and guiding decisions regarding its implementation and utilization within educational settings. Ultimately, the favorable evaluation outcomes affirm the system's efficacy in facilitating effective question generation and response, thereby contributing to enhanced user learning experiences and outcomes.

2.6 LLMCCE Pilot Study

To gauge the intelligent tutoring system's practical efficacy and user experience, access was extended to a select group comprising 8th-grade students and their dedicated history lesson human tutor. This endeavor aimed to solicit firsthand feedback from end users regarding their engagement with the system and overall satisfaction with its functionality and utility within the educational context.

During the evaluation period, students and their tutor interacted with the system, engaging in various activities designed to assess its effectiveness in facilitating learning and comprehension of historical concepts. Notably, participant feedback indicated a high level of engagement and enthusiasm with the system, with students expressing a keen interest in revisiting the procedures multiple times in pursuit of improved outcomes.

3 Limitations

The evaluation procedure for an intelligent tutoring system's question generation and response capabilities has limitations due to subjective evaluations from domain experts. Future research should include a larger pool of experts to improve the robustness and generalize ability of evaluation outcomes.

Moreover, the evaluation process in educational domains is influenced by contextual nuances, with the effectiveness of model question-answer pairs varying based on the context and instructional material complexity. Understanding these factors is crucial for making informed decisions about the model's applicability.

The model's relevance issue highlights the need for ongoing research and refinement of RAG approaches. Strategies like optimizing prompts or exploring alternative methodologies can enhance contextual relevance and accuracy, addressing evolving needs and expectations.

4 Conclusion

Our proposed methodology integrates natural language processing and interactive learning methodologies to enhance educational experiences, engaging learners, fostering active participation, and facilitating knowledge acquisition and retention. Simultaneously, an assessment is conducted by specialists in the field of LLM Model efficiency, namely the GPT-4 in our instance, concerning the task of identifying the salient features of the provided text and crafting focused questions around them.

Our system effectively delivers relevant and engaging educational content, with an intuitive interface and interactive features. However, further research is needed to address limitations, improve performance, and enhance user satisfaction. This will help the system evolve to meet diverse user needs and expectations.

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